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RAMS PLAN

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1. INTRODUCTION

1.1. General

This plan outlines BEML's approach to the implementation of RAMS (Reliability, Availability, Maintainability and Safety) approach for the management of operational reliability, availability, maintainability and safety during the design, development, production, operation & maintenance of the Rolling Stock.

RAMS activities are desirable for new railway systems to ensure the fundamental issues of Reliability, Availability, Maintainability and Safety. This plan is fully taken into account during the planning, design, construction, commissioning & operation and maintenance of the trains.

A variety of techniques have been utilized to enable a reasonable prediction of the likely performance of new trains in terms of reliability, availability and maintainability. During the design stages of a project, the benefits of such analyses can be significant in identifying potential problems early enough for action to be taken before manufacture or installation. At commissioning, reliability and availability demonstration activities are undertaken to validate the predictive and analytical techniques undertaken during the design. Objective of this plan is to assure that the reliability, availability and maintainability requirements of KMRC RS(3R) cars are satisfied.

This plan covers the RAM aspects in detail whereas the safety aspects are covered in System Safety Assurance Management Plan bearing No. GR/TD/3280, DRCA No.: RS(3R)/PP/P/B857/A1.

1.2. Rationale and Contributions

RAM Assurance activities are desirable for new railway system to ensure that the fundamental issues of Reliability, Availability, Maintainability (RAM) are fully taken into account in the planning, design, construction, commissioning, operation and maintenance of each system.

RAM Assurance encompasses the ongoing requirement to consider RAM through each stage of a project, from feasibility study through to commissioning and operation. A variety of techniques has been utilized to enable a reasonable prediction to be made of the likely performance of new railway in terms of RAM. If undertaken properly at feasibility study at the design stages of a Project, the benefits of such analyses can be significant in identifying potential problems early enough for action to

be taken before manufacture or installation. At commissioning, reliability, maintainability demonstration activities are undertaken to validate the predictive and analytical techniques adopted during the design. During commissioning, operation and defect liability period, verification of the predicted performance of the Rolling Stock in terms of the actual RAM will be carried out.

The RAM assurance activities are divided into the several stages. During design stages, BEML has identified the high level requirements for the train, and apportioned these requirements down to the individual sub-systems that the train comprises. Based on which a detailed design analyses and improvements are prepared. These analyses has supported design activities by highlighting any potential reliability problems in the design.

Reliability of the proposed design is being predicted at the conclusion of Final Design Completion. This demonstrates that the systems as built meet their design objectives. Finally, during Warranty Period, commissioning and operational experience during the defect liability period will allow an assessment of the actual reliability performance of the rolling stock in service.

1.3. Policy

BEML is committed to deliver rolling stock which meets the world-class standards of reliability, and which meets the specific requirements laid down in the KMRC RS(3R) project specification. This policy has been achieved through the implementation of a formal system assurance process facilitated by a System Safety and Reliability, Availability and Maintainability Management Plan. System assurance activities has been carried out in accordance with EN 50126-1999.

Though system assurance is a co-ordinated activity in support of rolling stock design, the responsibility for achieving world-class standards of reliability, availability and maintainability rests on all members of the rolling stock, in design, in manufacturing, in supply and procurement and in commissioning.

BEML's commitment to reliability and maintainability is cascaded down to our sub-contractors and suppliers, to ensure that we procure equipment to the high standards of reliability and maintainability that we expect from our own equipment. To support this objective, specific system assurance requirements and objectives have been set for each system and sub-system.

Reliability, Availability and Maintainability are the key qualities that the public expect

of KMRC's train operations, and BEML is committed to support KMRC meet that expectation.

2. References

This Plan makes reference to the following documents:

(1) Employer's Requirements : General Specification, KMRC

(2) Employer's Requirements : Technical Specification, KMRC

(3) EN50126:1999, Railway applications– The specification and demonstration of Reliability, Availability, Maintainability and Safety (RAMS)

3. Definitions

3.1. Glossary of Terms

These definitions are quoted from EN 50126:1999, Railway applications – The specification and demonstration of Reliability, Availability, Maintainability and Safety (RAMS)

Reliability	The probability that an item can perform a required function under given conditions for a given time interval; often expressed as Mean Time Between Failures (MTBF), assuming a constant failure rate.
Availability	The ability of a product to be in a state to perform its required function under given conditions at a given instant of time or over a given time interval assuming that the required external resources are provided.
Maintainability	The ease with which a system can be maintained, considering routine scheduled maintenance, unscheduled and breakdown maintenance.
Safety	Freedom from unacceptable risk or harm.
Risk	The probability or frequency of the realization of a Consequence causing a measure of loss.
Consequence	Condition which arises from the addition of energy, or exposure to a hazard. These may range from benign to catastrophic incidents and accidents. A range of consequences may be associated with a hazard.
Failure Mode	The predicted or observed results of a failure cause on a stated item in relation to the operating conditions at the time of the failure.
Failure Rate	The limit, if this exists, of the ratio of the conditional probability that the instant of time, T, of a failure of a product falls within a given time interval (t, t+Δt) and the

	length of this interval, Δt, when Δt tends towards zero, given that the item is in an up state at the start of the time interval.
Fault Mode	One of the possible states of a faulty product for a given required function.

3.2. Abbreviations and Acronyms

CM	Corrective Maintenance
PM	Preventive Maintenance
OPM	Other Preventive Maintenance
DT	Down Time
FDR	Final Design Review
FMEA	Failure Modes and Effects Analysis
FMECA	Failure Modes, Effects and Criticality Analysis
FRACAS	Failure Recording and Corrective Action System
FTA	Fault Tree Analysis
HL	Hazard Log
KMRC	Kolkata Metro Rail Corporation Limited
LRU	Line Replaceable Unit
MDBF	Mean Distance Between Failure
MDBCF	Mean Distance Between Component Failure
MTTR	Mean Time to Repair/Restore
PDR	Preliminary Design Review
RAMS	Reliability, Availability, Maintainability and Safety
SA	System Assurance
SAP	System Assurance Plan
SEM	System Engineering Management
SWG	System Assurance Cross Functional Working Group
TS	Employers Requirements – Technical Specification
TBD	To be determined

4. Scope of the RAM Management Plan

The first stage of the RAM assurance process was to define a clear plan for implementation of the RAM assurance activities. This has been achieved with reference to the KMRC Specification. This Plan will be the controlling document for RAM assurance activities for the KMRC project.

The Reliability, Availability and Maintainability Plan presents a program for RAM assurance activities, showing how they feed into the design, development and production process, to ensure that the Rolling Stock design achieves the required levels of RAM assurance in operation, and details the documentation and analyses that are generated to provide KMRC with assurance of this. The Reliability, Availability and Maintainability Plan identifies the responsibilities of the parties involved in the process with respect to RAM assurance and demonstrates to KMRC that the achievement of RAM assurance through design life cycle is being given sufficient priority during the design process.

The RAM assurance activities described in this plan applies to all aspects of rolling stock design and manufacture, including those systems and components to be constructed & integrated by BEML and various sub-contractors.

The system assurance organization presented in this plan shows how BEML Co-ordinates the RAM assurance activities of the various other parties involved in system design and manufacture.

The scope of RAM analysis is as follows:

- Propulsion System
- Auxiliary Supply System
- Air Supply and Friction Brake Equipment
- Door System and Controls
- VAC System
- Communication System
- Couplers and Draft Gear
- Bogies
- Lighting System
- TMS

5. Organization, Roles & Responsibilities

In particular, RAM activities are a part of system assurance. Therefore for the effective progress of system assurance including RAM activities, the system assurance specialist in organization structure is a skillful expert having experience of application of System Assurance process for previous projects.

6. Technical Requirements

Following are the technical requirements for Chapter. 19 : RAMS: Reliability, Availability, Maintainability, Safety of ERTS

ERTS Clause No	Requirements	Proposal
19.1	**Reliability, Availability and Maintainability: General**	
19.1.1	The Contractor shall submit Reliability, Availability and Maintainability Management Plan as specified in chapter 2 of the Employer‘s Requirements, General Specification. The Contractor shall verify, after system design have been completed, that the reliability, availability and maintainability requirement will be met.	Complied
19.1.2	The Contractor shall comply with the guidelines of IEC 60300-1, IEC 60300-2 and IEC 60571 for electronic equipment, and IEC 60300-3-5 in meeting the reliability, availability and maintainability requirements of equipment.	Complied. Covered under Section 3.2 of this document.
19.2	**Reliability Requirements**	
19.2.1	Kolkata Metro Rail Corporation attaches the greatest importance to the attainment of the highest possible Reliability during service of all the equipment and systems supplied and installed under this contract. The design, manufacture, installation and commissioning of the equipment as also the training of the operating and maintenance staff shall be such as to ensure near Zero Failure performance in the initial stages and that the few defects and deficiencies that may be exposed during the Service Trial and the initial reliability growth period of one year are totally eliminated in the bulk supply.	Noted

19.2.2	The Contractor shall demonstrate by quantitative methods achievement of the specified levels of reliability for the train and specific individual items of equipment.	Complied.. covered in section 7.2 of this doc
19.2.3	An evolving reliability model consisting of reliability block diagrams and probability of success equations shall be developed and submitted to the Engineer for acceptance. This model shall show the relationships required for system and equipment to operate successfully. The reliability block diagrams shall include all elements essential to the successful performance of the system and the interrelationships and interface of these elements.	Complied
19.2.4	Reliability apportionment and prediction analysis shall be in accordance with established techniques or standards, which will be submitted for acceptance by the Engineer. The analysis shall provide predictions for each major equipment and sub-system. Predictions shall be based on actual revenue service results for identical equipment operating under service conditions and duty cycles equivalent to KMRCL, or more severe. The analysis shall be carried out in parallel with the design of the train. The relevant apportionment and prediction figures shall be part of the design submission documents for the individual equipment, sub-system and system.	Complied
19.2.5	Reliability Apportionment and Prediction Report shall be completed prior to build commencing and reports shall be submitted at this stage for acceptance by the Engineer, who reserves the right to require the Contractor to carry out field data collection to verify the reliability model.	Complied
19.2.6	Definitions : (i) Relevant Failure : A relevant failure of an item is an independent failure which results in a loss of function of that item caused by any of the following: • A fault in an equipment or sub-system while operating within its design and environmental specification limits; • Improper operation, maintenance, or testing of the item as a result of the Contractor supplied documentation. • Failures of transient nature including those with post investigation status as ‗No fault found', shall be considered as relevant failure if in the opinion of the Engineer these are attributable to rolling stock. The decision of the Engineer shall be final. (ii) Non-relevant Failure: Any failure of an item not included in the definition of relevant failure, such as the following:	Noted. Covered in section 7.2 of this document

- A failure caused by malfunction of other equipment or subsystem that are not supplied by the Contractor;
- A failure caused by human error, except as noted in Relevant Failure above;
- A failure caused by accidents not associated with the normal operation of the item. Such as collision or striking a foreign object on the right of way;
- A failure caused by operating the equipment or sub-system outside of design or environmental specification limits.

(iii) Service Failure: Any relevant failure or combination of relevant failures during revenue service operations, simulated revenue operations, or during pre-departure equipment status checkouts to determine availability for revenue service, which results in one of the following:

- Non-availability of the train to start revenue service after successful completion of pre-departure checkout;
- Withdrawal of the train from revenue services;
- A delay equivalent to or exceeding 3 minutes.

The discretion of declaring a metro train as Not-available to start revenue service after successful completion of pre-departure checkout or withdrawing a train from revenue service on account of any relevant failure rests solely with the Employer and shall be final.

(iv) Pattern Failure: Repeated occurrence of three or more relevant failures of the same replaceable part, item or equipment in same manner in identical or equivalent applications when they occur at a rate, which is inconsistent with the predicted failure rate of the part, item or equipment. The percentage predicted rate shall be as under:

(a) three or more relevant Service Failures of the same replaceable part, item or equipment in same manner in identical or equivalent applications occur at a rate which is at least 20 % higher than the predicted failure rate of the part, item or equipment and / or

(b) at least 20 % of the same replaceable part, item or equipment in the fleet has a relevant failure in the same manner in identical or equivalent applications during a moving 18 months window starting when the reliability demonstration starts and ending at the end of the Defect Liability Period, these failures will be classed as pattern failures.

(v) Mean Distance Between Failure (MDBF): The MDBF is the ratio of the total operating distance accumulated by the total available fleet of the trains to the total number of Service Failures

(vi) Mean Distance Between Component Failures (MDBCF): The MDBCF of a system is the ratio of the total operating distance accumulated by the total population of

	identical items in the available fleet of the trains to the total number of Service failures occurring within the population identical items.	
19.2.7	Reliability Targets: (i) Reliability shall be monitored for the respective fleet for 6-car metro trains. (ii) The fleet average levels of MDBF, as specified in table 19.1, shall be achieved.	Complied
	Table 19.1: Reliability Targets Duration: After 6 months of start of revenue service — Minimum fleet average MDBF 6-Car fleet: 60,000 Duration: After 12 months of start of revenue service — Minimum fleet average MDBF 6-Car fleet: 100,000	Complied
	Any train shall be counted as available for reliability calculations only after a stabilization period of 6 months after putting the train into revenue service. (iii) The achieved level of MDBCF of major systems shall be as proposed by the Contractor in the bid. The Tenderer shall submit MDBCF of the major systems as listed in table 19.2 along with the bid. **Table 19.2: MDBCF of major systems** S.N: (i) System / Equipment: Propulsion System — MDBCF (km): a) Shoe collectors — b) HSCB and Earthing switch — c) Traction Inverter — d) Traction Motor — (ii) Auxiliary Supply System — a) Auxiliary Converter — b) Battery Charger — c) Back-up Batteries — (iii) Air Supply and Friction Brake Equipment — (iv) Door System and Controls — (v) VAC System — (vi) Communication System — (vii) Couplers and Draft Gear — a) Automatic couplers — b) Semi permanent couplers — (viii) Bogies — a) Drive gear and coupling — b) Primary suspension — c) Secondary suspension — (ix) Lighting System — (x) DMS —	Complied
19.3.	**Reliability Demonstration**	
19.3.1	The Contractor shall be required to establish a personal computer based Failure Reporting and Corrective Action	Complied.covered in section 9.4.2 of this

	(FRACAS) System to demonstrate compliance with specified train and equipment reliability. The reliability demonstration of each train shall start after six months of that train in revenue service and shall continue till the end of the defects liability period. Reliability of the trains and of the identified major systems shall be demonstrated on fleet basis.	document
19.3.2	The Employer shall collect and maintain data on every Service Failure along with the TMS data indicating the probable failure. MDBF and MDBCF shall be calculated throughout the monitoring period. The Contractor shall collect all the relevant details from the Engineer and submit monthly Reliability Demonstration Reports.	Complied
19.3.3	In case the Contractor is not able to achieve specified/provided reliability target of MDBF/MDBCF, the Contractor shall take necessary corrective measures either by way of change of design of the relevant equipment/ component or software modification.	Complied. Covered under section 7.4 of this document
19.3.4	The Contractor shall analyze each and every failure/defect of components of various equipments to determine the cause of failure and to propose corrective measures, which would be reviewed by the Engineer.	Complied
19.3.5	A record shall be maintained for each and every defect/failure in accordance with FRACAS to be submitted by the Contractor and approved by the Engineer.	Complied
19.3.6	Reliability shall be monitored during revenue service operation of the trains. The Contractor shall collect and collate data on each and every deficiency and failure observed by both himself and the Engineer, from handing over the first train to the end of the Defect Liability Period. Each and every failure, whether of component, sub-system or system, during this period shall be subject to a failure analysis to determine the cause of failure? The Contractor shall submit investigation reports for review of the Engineer.	Complied. Covered under Section 7.4 of this document
19.3.7	Correction shall be made to components or subsystems that either fail to attain predicted reliability levels or show Pattern Failure, at no additional cost to the Employer.	Complied
19.4	**Availability Requirements**	
19.4.1	Availability shall be assessed by the following measure:	Noted. Covered under section

	Percentage Availability = $\left[\frac{1-[DT(OPM)+DT(CM)]}{TotalTime}\right]\times$ Where: (i) Total Time is the time in hours in the assessment period multiplied by the number of trains commissioned under the Contract. (ii) DT (OPM), or Down Time due to Other Preventive Maintenance, is the total down time in hours due to Preventive Maintenance other than service checks, summed over all sessions carried out on all trains commissioned under the Contract during the assessment period. The trains shall not be due for major overhauls at the time of demonstration and shall therefore be excluded from the assessment. (iii) DT (CM), or Down Time due to Corrective Maintenance, is the total down time in hours due to corrective maintenance, summed over all sessions carried out on the trains commissioned under the Contract during the assessment period. Any unreasonable delay in handing-over the train for repairs for reasons not attributable to contractor shall be excluded. Time spent on train integrity inspections after train reformations arising from corrective maintenance work shall be included. (iv) The down time DT (OPM) shall be counted starting from the moment when the train becomes unfit for service or work is physically started on a train, whichever is earlier, and shall end when the train is restored to service condition. If the train is withdrawn from revenue service specially for preventive maintenance, time spent on withdrawing the train and sending back the train to revenue service, if any, shall also be included. (v) Down time DT (CM) shall be counted starting from the moment when the train becomes unfit for service or work is physically started on a train, whichever is earlier, and shall end when the train is restored to service condition. If the train is sent to revenue service after the corrective maintenance, the time spent on sending back the train to revenue service, if any, shall also be included. (vi) The down times DT (OPM) and DT (CM) shall also cover the full content of the maintenance work concerned, including safety precautions, inspections, servicing, replacement of equipment, defect detection and rectification, testing and restoration to service condition.	8.2 of this document
19.4.2	Availability Target: the trains supplied shall achieve a minimum availability of 95%.	Complied. Covered under section 8.2 of this document
19.5	**Availability Demonstration**	
19.5.1	The average availability of the trains shall be assessed after	Complied. Covered under

	18 months from the start of revenue operation with the first train supplied under the contract, in a specified train Maintenance Depot. The total maintenance down times for all trains shall be collected by the Engineer on monthly basis, and the average availability during the preceding six months, shall be worked out from the above formula.	section 8.2 of this document
19.5.2	In the event that the availability target is not achieved due to Rolling Stock, the determination of availability achievement in the preceding six month period shall be continued at monthly intervals until the target is achieved.	Complied. Covered under section 8.2 of this document
19.5.3	In the event that the availability target is not achieved, the Contractor shall, at his own expense, take whatever action is deemed necessary to meet the availability requirement.	Complied. Covered under section 8.2 of this document
19.6	**Maintainability Requirements**	
19.6.1	Simplicity of maintenance, operation and emergency procedures, ease of repair of damaged cars and equipment, are most important. These together with ease of exterior and interior cleaning will be taken into account throughout the development of the design.	Complied. Covered under section 8.3 of this document.
19.6.2	Particular attention shall be paid during the design of the cars to ensure that scheduled maintenance tasks are achieved in minimum time and using minimum manpower.	Complied. Covered under section 8.3 of this document.
19.6.3	Those components, systems and assemblies which require routine maintenance, frequent attention or unit replacement, shall be easily accessible for in situ maintenance.	Complied. Covered under section 8.3 of this document.
19.6.4	The Contractor shall develop a comprehensive maintenance programme for the trains.	Complied. Covered under section 8.3 of this document.
19.6.5	The maintenance regime proposed for the train shall be developed during the design process. A Failure Mode Effect Analysis (FMEA) will be required, based on function and derived from the specification at conceptual design stage.	Complied. Covered under section 8.3 of this document.
19.6.6	At pre-final design stage the Contractor will develop this FMEA to include required maintenance derived from each failure mode. Any other maintenance required for the train should be indicated at this stage.	Complied. Covered under section 8.3 of this document.
19.6.7		Complied.

	The vehicle shall incorporate design, which reduces maintenance, substantially improving service intervals and component replacement. The design shall also minimize mean time to repair (MTTR) and costs throughout design life. MTTR is the ratio of cumulative time, including the access time expended during a time interval to the total number of relevant failures.	Covered under section 8.3 of this document.
19.6.8	The objective of the maintainability program including corrective and preventive maintenance shall provide for: (i) Enhancement of Vehicle availability. (ii) Minimisation of maintenance cost. (iii) Minimisation of vehicle down time.	Complied. Covered under section 8.3.2
19.6.9	During the design stage, the Contractor shall furnish a list of Least Replaceable Units (LRU's) for the equipments, Sub-system and Systems supplied, which should not take more than 30 minutes for replacement. Specific exceptions, if any, whose replacement is not achievable in 30 minutes, shall be indicated by the tenderers in their offer. In order to achieve this requirement, quick release connections such as plugs and adaptor shall be provided between LRU's and the equipment.	Complied
19.6.10	The Tenderer shall submit the expected MTTR of the identified key systems as listed in table 19.6, along with the bid.	Complied

Table 19.6: MTTR of major systems

S.N	System / Equipment	MTTR (hours) Target
(i)	Propulsion System	2.5
	a) Third rail current collector	1.5
	b) HSCB & Earthing switch	2.5
	c) Power Converter – Inverter	3.0
	d) Traction Motor	3.0
(ii)	Auxiliary Supply System	1.7
	a) Auxiliary Converter	2.0
	b) Battery Charger	2.0
	c) Back-up Batteries	1.0
(iii)	Air Supply and Friction Brake Equipment	2.0
(iv)	Door System and Controls	1.2
(v)	VAC System	1.5
(vi)	Communication System	1.2
(vii)	Couplers and Draft Gear	2.2
	a) Automatic couplers	2.2

S.N	System / Equipment	MTTR (hours) Target
	b) Semi permanent couplers	2.0
(viii)	Bogies	2.0
	a) Drive gear and coupling	2.0
	b) Primary suspension	2.0
	c) Mechanical Parts & Components	2.0
	d) Secondary suspension	2.0
(ix)	Lighting System	1.0
(x)	DMS	1.5

19.7	**Maintainability Demonstration**	
19.7.1	The Contractor shall carry out tests to demonstrate that all maintainability predictions provided are met. All such tests shall be completed within six months after the delivery of first train. In the event that any maintainability target is not achieved, the Contractor shall at his own expense take whatever action is deemed necessary to meet the maintainability targets.	Complied. Covered under section 9.3 of this document
19.7.2	The Contractor shall ensure that all the required information including the related Maintenance Work Instructions (MWI) etc. are available before the commissioning of the first train into revenue service to enable him to demonstrate the maintainability targets.	
19.8	**Vehicle Maintenance Program**	
19.8.1	The Contractor shall submit to the Engineer a maintenance program, within the period specified in the delivery schedule, in order to reach the reliability and availability level defined in the performance requirements.	Complied. Covered under O&M manuals.
19.8.2	During the design stage, the Contractor shall submit downtime and manpower requirements for the maintenance inspections and service checks considered necessary for maintaining the trains under normal operational conditions as per table 19.7. The service check sessions shall include all routine maintenance activities including inspections, cleaning, washing, pest and rodent control etc.	Complied. Covered under O&M manuals.

<table>
<tr><td></td><td>Table 19.7: Maintenance Inspections and Service checks
<table>
<tr><th rowspan="2">Session</th><th rowspan="2">Interval (Minimum)</th><th colspan="2">Manpower and downtime requirements (Maximum)</th></tr>
<tr><th>Downtime</th><th>Expected staff (man days)</th></tr>
<tr><td>Service Check 1</td><td></td><td></td><td></td></tr>
<tr><td>Service Check 2, if any</td><td></td><td></td><td></td></tr>
<tr><td>Service Check 3, if any</td><td></td><td></td><td></td></tr>
<tr><td>.....</td><td></td><td></td><td></td></tr>
<tr><td>Service Check n, if any</td><td></td><td></td><td></td></tr>
</table></td><td>Complied. Covered under O&M manuals.</td></tr>
<tr><td>19.8.3</td><td>The Contractor shall also submit periodicity, downtime and manpower requirements for the maintenance activities as listed in table 19.8, for maintaining the trains under normal operational conditions, during the design stage.</td><td>Complied. Covered under O&M manuals.</td></tr>
<tr><td>19.8.4</td><td>For achieving lowest operation and maintenance costs, the metro trains shall be designed according to the following criteria:
(i) Design for a life cycle of at least 35 years
(ii) Main mid life overhaul after 15 years
(iii) Interval for Revision / main inspection of traction and brake equipment and other safety relevant equipment about 500 000 km</td><td>Complied. Covered under O&M manuals.</td></tr>
<tr><td></td><td>Table 19.8: Maintenance Activities
<table>
<tr><th>Type of activity</th><th>Frequency (days or km)</th><th>Expected staff (man days)</th><th>Downtime</th><th>Specific tools required</th></tr>
<tr><td>A</td><td></td><td></td><td></td><td></td></tr>
<tr><td>B</td><td></td><td></td><td></td><td></td></tr>
<tr><td>C</td><td></td><td></td><td></td><td></td></tr>
<tr><td>...</td><td></td><td></td><td></td><td></td></tr>
<tr><td>Half-life overhaul</td><td></td><td></td><td></td><td></td></tr>
</table></td><td>Complied. Covered under O&M manuals.</td></tr>
<tr><td>19.8.5</td><td>The Corrective Maintenance time as indicated above shall include defect identification, replacement of defective LRUs and restoration to service condition</td><td>Complied. Covered under O&M manuals.</td></tr>
<tr><td>19.8.6</td><td>For all the life of the metro train, this plan shall include also :
(i) maintenance concept (schedule, type of operation…),
(ii) maintenance documentation,
(iii) labour and infrastructure means,
(iv) tools and workshop equipment,
(v) spare parts (including wearing parts),</td><td>Complied. Covered under O&M manuals.</td></tr>
<tr><td>19.8.7</td><td>The trains shall operate with minimum attention between the specified inspection periods, and shall, under the operating conditions specified, operate between overhaul</td><td>Complied. Covered under O&M manuals.</td></tr>
</table>

	periods without requiring replacement of components other than those on the agreed list of consumable parts to be proposed by the Contractor and accepted by the Engineer.	
19.8.8	Special tools shall be avoided wherever possible. If they are required, they shall be supplied by the Contractor in sufficient quantities to meet the maintenance requirements.	Complied. Covered under O&M manuals.
19.8.9	Equipment design shall be modular to minimise down time following failures of equipment and components. Provision for mechanical handling devices shall be provided for any single piece of equipment weighing more than 35kg. Equipment covers shall be provided with secure, visible, latching arrangements for easy inspection from the side of trains	Complied. Covered under O&M manuals.
19.8.10	All under frame equipment which cannot be handled manually shall be configured such that it can be removed and replaced from track level using fork lift trucks or lift tables, with recognition being given to the confined environment of the pit and the rail level and under frame dimensions. All under frame equipment shall be arranged such that it is capable of being removed and replaced without disturbing any other equipment.	Complied. Covered under O&M manuals.
19.8.11	If any equipment mounted above the ceiling requires the use of lifting equipment for its removal or refitting this shall be readily achievable without the risk of damage to the vehicle interior.	Complied. Covered under O&M manuals.
19.8.12	Removal and re-assembly of moving and wearing parts on bogies shall generally be carried out without the use of special tools.	Complied. Covered under O&M manuals.
19.8.13	Bogies shall be capable of being disconnected and reconnected to vehicle bodies with a minimum of operations. All connections must be easily and safely accessible to personnel located in pits or alongside the bogie at rail level. It shall be easy to inspect for correct reconnection, from alongside the bogie where possible. Preference will be given to a design which permits release of the bogie to permit the raising of the car body, without the need for a pit in the Lifting Berth.	Complied. Covered under O&M manuals.
19.8.14	Each vehicle shall be capable of being lifted complete with bogies without the need to attach extra restraints or	Complied. Covered under O&M manuals.

	supports for the bogies or wheels.	
19.8.15	Lubrication points shall have button head type grease nipples, and shall be easily accessible from rail level and shall, where possible, be grouped together.	Complied. Covered under O&M manuals.
19.8.16	On-vehicle test equipment shall be used on a vehicle to discriminate between a fault on the main equipment and a fault on the control electronic equipment.	Complied. Covered under O&M manuals.
19.8.17	Should the electronic equipment be found to be faulty, the equipment shall enable fault finding to be carried out at module level.	Complied. Covered under O&M manuals.
19.8.18	Off vehicle test equipment shall be used in the depot repair centre. This equipment shall allow fault finding down to the smallest replaceable item of equipment.	Complied. Covered under O&M manuals.
19.8.19	The unit shall have equipment cases and modules that are connected to the main vehicle wiring via connectors which are proven in equivalent service duties to achieve high reliability and are easily removable in the event of equipment replacement.	Complied. Covered under O&M manuals.
19.8.20	Equipment to which access will be required for faultfinding shall be conveniently located. A list of such equipments and their location shall be supplied.	Complied. Covered under O&M manuals.
19.8.21	The unit shall have provision for the isolation and where applicable, earthing of all electrical sub-systems to facilitate safe and systematic maintenance and fault diagnosis.	Complied. Covered under O&M manuals.
19.8.22	It shall be physically impossible for plug and socket connections and connections on safety-critical circuits to be mismatched.	Complied. Covered under O&M manuals.
19.8.23	The unit shall have standard test points on pneumatic systems. There shall be unrestricted access to facilitate checks during routine maintenance and fault diagnosis.	Complied. Covered under O&M manuals.
19.8.24	The above mentioned features shall be suitable reflected in the respective design documents, as applicable, during the design stage.	Complied. Covered under O&M manuals.

19.9	**Safety Assurance**	
19.9.1	Safety is defined as freedom from those conditions that can cause death, injury, occupational illness, or damage to or loss of equipment or property, or withdraw the train from service. Is considered as a risk all circumstance susceptible to cause injuries or person death (passengers, operation staff, maintenance staff), and by extension all event leading to a partial or total destruction of costly equipments. The objective of safety is expressed by" the capacity of the rolling material to keep the physical integrity of the passengers and persons in general ". The safety of the metro train aims to reduce to an acceptable level the probability of occurrence of catastrophic and critical events (according to norm EN 50126-1).	Complied. Covered in Doc No: GR/TD/3280
19.9.2	The Contractor shall submit a “System Safety Assurance Management Plan‘ for review and acceptance by the Engineer as specified in the Engineer‘s Requirements: General Specification chapter 2.	Complied. Covered in Doc No: GR/TD/3280
19.9.3	The System Safety Assurance Management Plan shall cover design, manufacture, testing, commissioning and integrated testing, and minimising the magnitude and seriousness of events or malfunctions, which could result in injury to patrons or staff and damage to equipment or property, but cannot be completely eliminated.	Complied. Covered in Doc No: GR/TD/3280
19.9.4	All equipment and systems, including software, affecting train safety and the safety of train crew and passengers, and/or identified as being —vital, shall be designed according to the following principles: (i) Only components having a high reliability and predictable failure mode shall be used. (ii) Components must be utilized in such a manner that ensures that a restrictive, rather than a permissive condition will result from a component failure. (For example: brakes will apply, rather than release; train will decelerate, rather than accelerate.). (iii) Circuits shall be designed such that when a normally energized electric circuit is interrupted or de-energized, it will cause the controlled function to assume its most restrictive condition. (Broken wires, damaged or dirty contacts, a relay failing to respond when energized, etc., shall not result in an unsafe condition.) (iv) System safety equipment design must be such that any single independent component or subsystem failure results in a restrictive condition. Failures that are not independent, those failures which, in turn, always cause others, must be	Complied. Covered in Doc No: GR/TD/3280

	considered in combination as a single failure and must not cause a permissive condition.	
19.9.5	During the Design Review process, the Contractor shall submit analyses for Engineer review, which demonstrate compliance with these safety principles. These analyses shall address the following issues: (i) Circuit design. (ii) Hardware design (Failure Modes, Effect and Criticality Analysis). (iii) Electrical interference. (iv) Software errors. (v) System failures.	Complied. Covered in Doc No: GR/TD/3280
19.10	**Hazard Analysis report**	
19.10.1	Safety performance shall be assessed by the Risk levels of individual hazards identified during the System Assurance process.	Complied. Covered in Doc No: GR/TD/3619
19.10.2	The Contractor shall submit the Hazard Analysis report in accordance with the requirements of Chapter 2 of the Employer‘s Requirements – general Specification and shall evaluate and ensure that all the hazards are identified and satisfactorily resolved.	Complied. Covered in Doc No: GR/TD/3619
19.10.3	The Contractor shall take lead role in the interface Hazard Analysis for train borne equipment provided by other contractors.	Complied. Covered in Doc No: GR/TD/3619
19.10.4	The Contractor shall produce the Hazard Analysis Schedule for the complete train including all train borne systems and shall interface principally with the Signalling, Communication, Power Supply, Civil and Depot Contractor as well as any other Designated Contractors to obtain the information necessary, from their hazard analysis, to complete the analysis.	Complied. Covered in Doc No: GR/TD/3619
19.10.5	The Contractor shall, as part of the safety analysis, prepare analysis to identify Hazards and ensure their satisfactory resolution. The following analysis shall be prepared and submitted by the Contractor for the Engineer acceptance. (i) Preliminary hazard analysis (ii) Interface hazard analysis (excluding EMI) (iii) Subsystem hazard analysis (iv) Operating hazard analysis including maintenance (v) Quantitative fault tree analysis (vi) Failure modes effects and criticality analysis (FMECA)	Complied. Covered in Doc No: GR/TD/3619

19.10.6	The Hazard Analysis shall be carried out in accordance with EN50126-1 as the primary standard, or any other internationally accepted equivalent standard, in areas not adequately addressed by the former standard.	Complied. Covered in Doc No: GR/TD/3619
19.10.7	The Contractor shall compile a list of critical and catastrophic items identified as a result of hazard analysis, FMECA or by other means. The Contractor shall carryout the Hazard and FMECA for the following equipment / sub-systems / systems: (i) Bogie and Suspension. (ii) Vehicle Body. (iii) Transmission Drive System. (iv) Gangways. (v) Coupler. (vi) Brake System. (vii) Door System. (viii) VAC System. (ix) Pneumatic System. (x) Communication System. (xi) HV and Propulsion System. (xii) Auxiliary Power System. (xiii) Control equipment. (xiv) TMS.	Complied. Covered in Doc No: GR/TD/3619
19.10.8	All hazard resolution by procedural control shall be cross-referenced from the Critical and Catastrophic Items List to the appropriate manuals.	Complied. Covered in Doc No: GR/TD/3619
19.10.9	As example, and according to the probability of occurrence associated to the event, these objectives are The qualitative measures of hazard severity are defined as follows: (i) Hazard Category I – Catastrophic: Operating conditions such that personnel errors, environment, design deficiencies, subsystem or component failure or procedural deficiencies may cause death or system loss. The safety target shall be based on internationally accepted standards. (ii) Hazard Category II – Critical: Operating conditions such that personnel errors, environment, design deficiencies, subsystem or component failure or procedural deficiencies may cause severe injury to personnel, severe occupational illness or major system damage. The safety target for the occurrence of all Category II hazards summed together shall again be based on internationally accepted standards. (iii) Hazard Category III – Marginal: Operating conditions such that personnel errors, environment, design	Complied. Covered in Doc No: GR/TD/3619

	deficiencies, subsystem or component failure or procedural deficiencies, may cause minor injury to personnel, minor occupational illness or minor system damage. (iv) Hazard Category IV – Negligible: Operating conditions such that personnel errors, environment, design deficiencies, subsystem or component failure or procedural deficiencies will not result in injury to personnel occupational illness or damage to the system. (v) The Contractor shall submit a Schedule for Hazard Analysis Submissions within 30 days of Notice to Proceed(NTP) and the Preliminary Hazard Analysis shall be submitted within 6 months of Notice to Proceed(NTP). This draft shall include a comprehensive assessment of potential equipment failure modes during normal operating and overload conditions and assess the performance of the equipment for a range of hazard conditions. The final draft shall be submitted by the completion date of final design.	
19.10.10	The procedures for Operation, Maintenance, Training and the Contractor‘s Quality Assurance manuals shall incorporate resolution of hazards so identified from this hazard analysis. Proper cross-referencing to the hazards and resolution measures shall be provided in all these aforementioned documents.	Complied. Covered in Doc No: GR/TD/3619
19.10.11	The following targets norms shall be employed for the Fault Tree Analysis These norms are subject to review by the Engineer during the detailed design stage, and mutually agreed upon. (i) No single point failure shall lead to fatality. (ii) No combination of undetected failure and double point failures shall result in fatality. (iii) No combination of undetected failure and single point failure shall result in major injury.	Complied. Covered in Doc No: GR/TD/3619
19.10.12	Source of all failure rates employed to be indicated in the Hazard Analysis.	Complied. Covered in Doc No: GR/TD/3619
19.10.13	All hazard analyses submitted to the Engineer are to be standardized by the Contractor such that format and forms employed by all sub-contractors are the same.	Complied. Covered in Doc No: GR/TD/3619
19.11	**Fire Safety Analysis Report**	
19.11.1	The Contractor shall prepare a Fire Safety analysis report for review and acceptance by the Engineer as required in chapter 3 of the Employer‘s Requirements - General Specification. Materials used in the cars shall conform to fire safety requirements of BS 6853: 1999 and NF F 16 101,	Complied. Covered in Doc No: GR/TD/3620

	NF F 16 102 or the latest edition of other equivalent international standards, subject to the acceptance of the Engineer.	
19.11.2	Whichever Standard is selected for meeting the Fire Safety Criteria, then that standard shall be declared, and once accepted by the Engineer, its requirements shall be met consistently throughout.	Complied. Covered in Doc No: GR/TD/3620

7. Reliability, Availability, Maintainability (RAM)

7.1. Purpose

This document outlines the process that has been followed to ensure that the KMRC rolling stock meets its RAM objectives.

7.2. Procedure

RAM Procedure is illustrated in Figure 1. The RAM Procedure is based on the achievement of quantitative reliability objectives and the fulfillment of qualitative availability and maintainability requirements.

The failure definition for reliability analysis as per specification is as follows:

(i) Relevant Failure : A relevant failure of an item is an independent failure which results in a loss of function of that item caused by any of the following:

- A fault in an equipment or sub-system while operating within its design and environmental specification limits;
- Improper operation, maintenance, or testing of the item as a result of the Contractor supplied documentation.

Failures of transient nature including those with post investigation status as 'No fault found', shall be considered as relevant failure if in the opinion of the Engineer these are attributable to rolling stock. The decision of the Engineer will be considered as final.

(ii) Non-relevant Failure: Any failure of an item not included in the definition of relevant failure, such as the following:

- A failure caused by malfunction of other equipment or subsystem that are not supplied by the Contractor;
- A failure caused by human error, except as noted in Relevant Failure above;

- A failure caused by accidents not associated with the normal operation of the item. Such as collision or striking a foreign object on the way;
- A failure caused by operating the equipment or sub-system outside of design or environmental specification limits.

(iii) Service Failure: Any relevant failure or combination of relevant failures during revenue service operations, simulated revenue operations, or during pre-departure equipment status checkouts to determine availability for revenue service, which results in one of the following:

- Non-availability of the train to start revenue service after successful completion of pre-departure checkout;
- Withdrawal of the train from revenue services;
- A delay equivalent to or exceeding 3 minutes from the Schedule / Time table as noted at the destination station for the one way trip.

The discretion of declaring a train as Not-available to start revenue service after successful completion of pre-departure checkout or withdrawing a train from revenue service on account of any relevant failure rests solely with the Employer and will be considered as final.

(iv) Pattern Failure: Repeated occurrence of three or more relevant failures of the same replaceable part, item or equipment in same manner in identical or equivalent applications when they occur at a rate which is inconsistent with the predicted failure rate of the part, item or equipment. The decision of the Engineer will be considered as final. The percentage predicted rate will be as under:

(a) three or more relevant Service Failures of the same replaceable part, item or equipment in same manner in identical or equivalent applications occur at a rate which is at least 20% higher than the predicted failure rate of the part, item or equipment and / or

(b) at least 20% of the same replaceable part, item or equipment in the fleet has a relevant failure in the same manner in identical or equivalent applications during a moving 18 months window starting when the reliability demonstration starts and ending at the end of the Defect Liability Period, these failures will be classed as pattern failures.

(v) Mean Distance Between Failure (MDBF): The MDBF is the ratio of the total operating distance accumulated by the total available fleet of the trains to the total number of Service Failures.

(vi) Mean Distance Between Component Failure (MDBCF): The MDBCF of a system is the ratio of the total operating distance accumulated by the total population of identical items in the available fleet of the trains to the total number of Service failures occurring within the population of identical items.

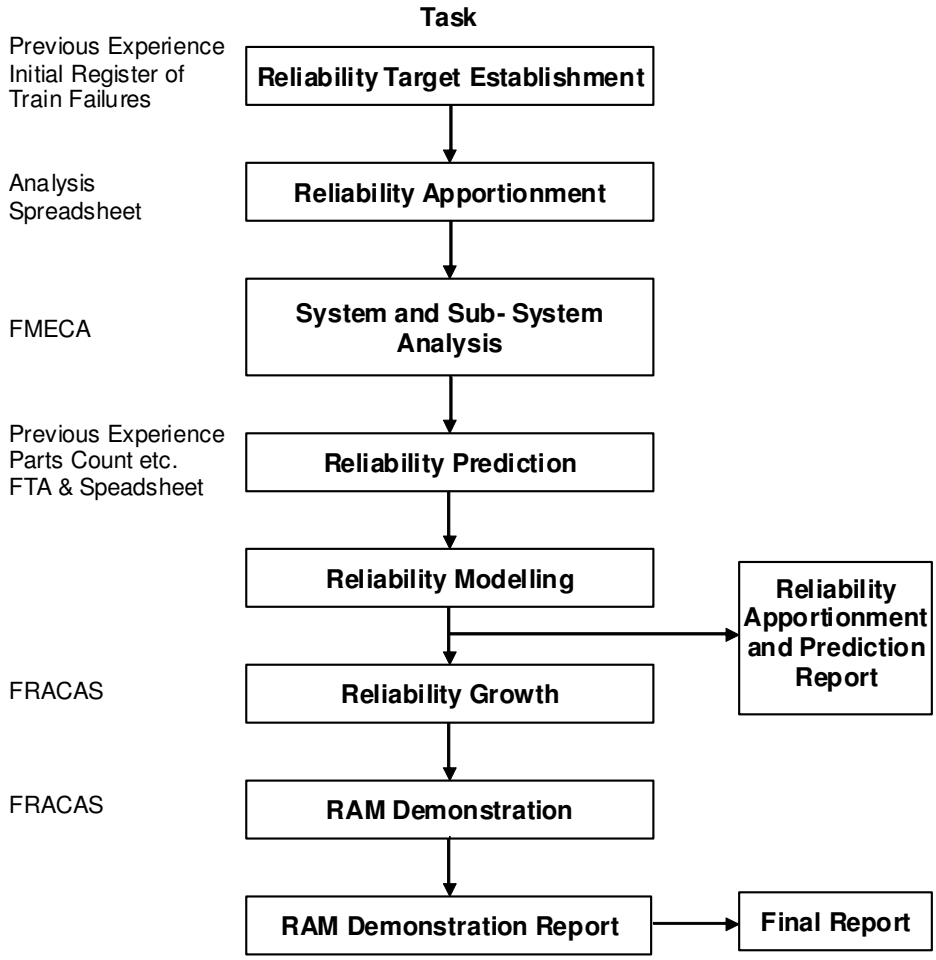


Figure 1: RAM Procedure

Apportionment of reliability targets to the system and sub-system level is based on the overall reliability objectives for the train as a whole. Specific maintainability requirements are also proposed for each system and sub-system. Reliability targets are apportioned for all train failures.

It is intended that failures resulting in a delay of three minutes or more will be

eliminated as far as possible. Each sub-system and system will then be subject to Failure Mode Effects and Criticality Analysis. The level of detail to which the analysis will be carried out will commensurate with the criticality of the system. Systems for which the failure could result in fatality or major injury, will be reviewed to a level of detail to identify the safety-critical components, or to confirm that appropriate safeguards are in place to prevent single-point failure resulting in injury or death.

Reliability prediction is based on previous operating experience of similar equipment, and/or vendor provided data. If this is not available, analytical techniques such as parts count analysis based on MIL STD 217F or equivalent international standards and data sources are applied. Reliability predictions has been carried out at the system level, and apportioned between system failure modes as required in order to support the overall rolling stock reliability prediction. Where the consequences of different failure modes are similar (e.g. several failure modes resulting in withdrawal of a train from service and replacement of the affected equipment) there will be no need to apportion reliability predictions between the various failure modes.

The reliability model is built based on the target apportionment. This reliability model has been used to combine the reliability predictions for individual systems and sub-systems into an overall train reliability performance prediction. Fault tree analyses has been used to model the likelihood of critical failures resulting from faults affecting multiple failures (e.g. failure of one auxiliary converter and failure of one air conditioning unit, neither of which individually would result in train failure, but, in combination, could result in train failure). The reliability model will provide KMRC with assurance that the rolling stock can meet its reliability objectives.

A failure recording and corrective action system will be set up to monitor the results of system test and demonstration, and to monitor the reliability performance of the rolling stock during the Defect Liability Period. During this period, any unacceptable systematic weaknesses identified will be managed through design improvement and modification to the rolling stock, in order to achieve the overall reliability objectives.

In particular, the electronic equipment subcontractor is requested to comply with

the guidelines of IEC 60300-1, IEC 60300-2, IEC 60300-3-5 and IEC 60571.

7.3. Results and Documentation

The reliability apportionment and prediction analyses prepared during the RAM analysis process including the result of availability calculation is covered in this document.

7.4. Responsibility

The safety organization is covered in detail in System Safety Assurance Management Plan bearing No. GR/TD/3280, DRCA No.: RS(3R)/PP/P/B857/A1. System Assurance Manager will be responsible for ensuring that all reliability engineering activities are carried out. He will ensure that all systems are subject to failure modes, effects and criticality analysis to an appropriate level of detail, and their contribution to rolling stock reliability estimated, based on valid reliability predictions.

The System Assurance Manager will be responsible for ensuring that the rolling stock reliability model spreadsheet is maintained up-to-date, based on data submitted by the various sub-contractors / suppliers.

The System Engineering Management Representative will be responsible for endorsing the system assurance analyses prior to their submission to KMRC for approval. He will be responsible for approving the targets specified for each sub-contractors systems and sub-systems, and for approving any variation to those targets.

Reliability shall be monitored during revenue service operation of the trains. The SEM Representative shall collect and collate data on each and every deficiency and failure observed by both himself and the Engineer, from handing over the first train to the end of the Defect Liability Period. Each and every failure, whether of component, sub-system or system, during this period shall be subject to a failure analysis to determine the cause of failure. The Contractor shall submit investigation reports for review of the Engineer.

7.5. Approach to RAM Assessment

BEML's approach to RAM assessment is shown in Figure 2 below. This shows the interrelationship between each RAM analysis

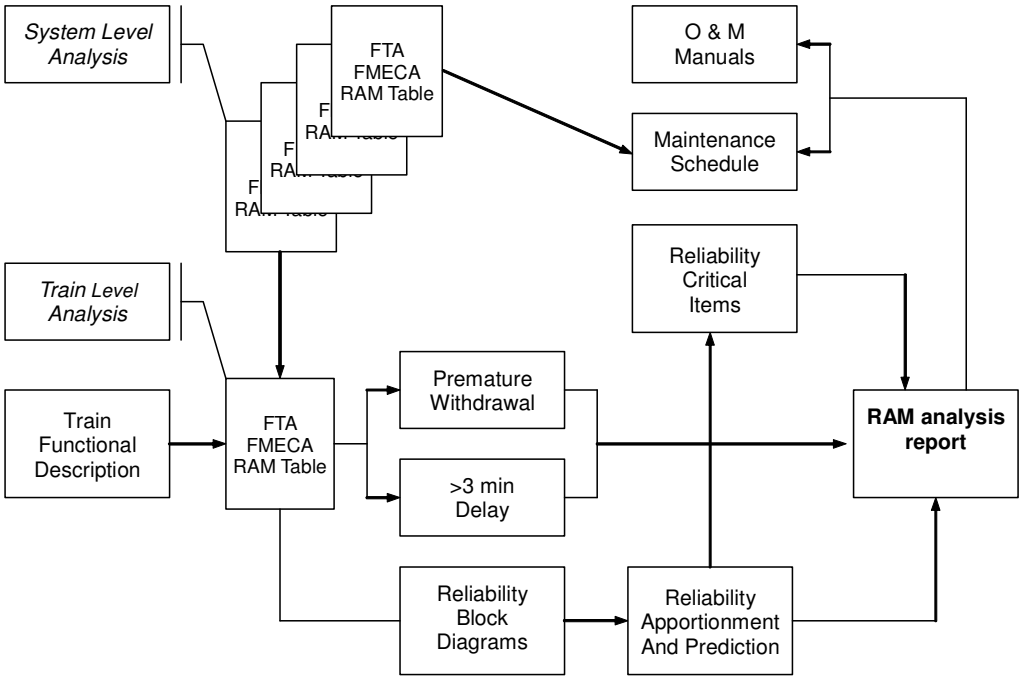


Figure 2: BEML Approach to RAM Assessment

7.6. Supplier management

BEML's approach to Sub-supplier management is shown in Figure 3.

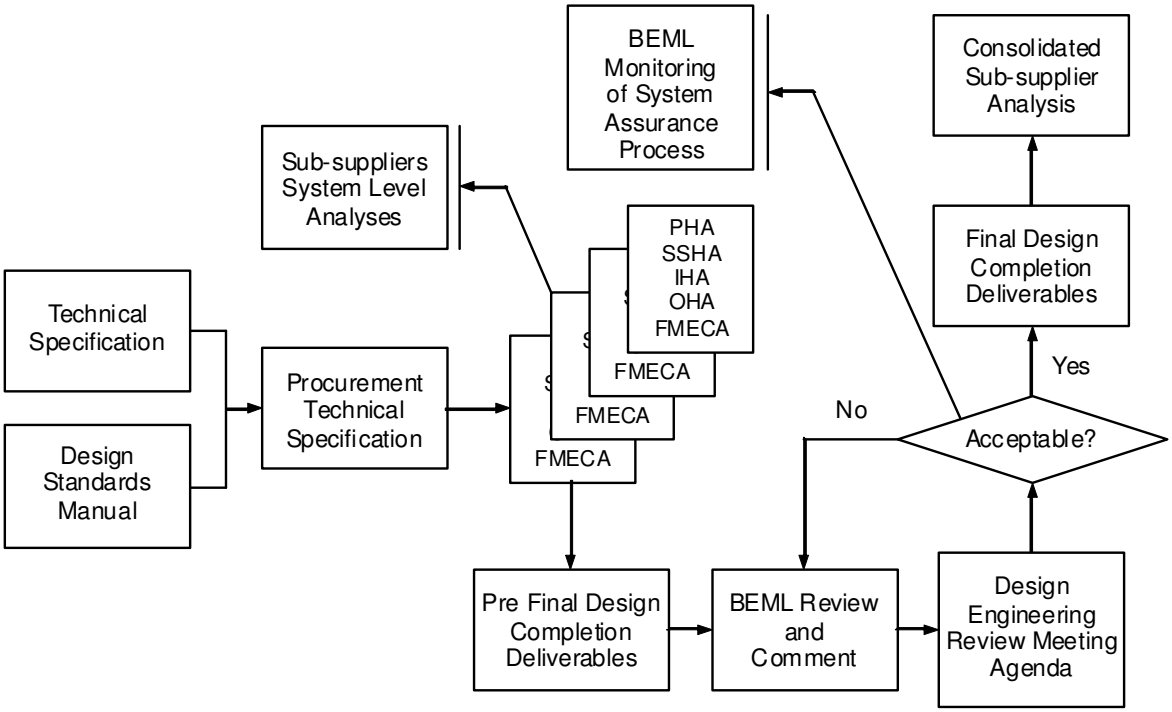


Figure 3: BEML's Approach to Sub-supplier Management

Supplier's RAMS requirements which will be included in Procurement Technical Specification (PTS) will be developed based on TS and Design Standard Manual. The required deliverable lists and expected date are stated in the contract to ensure RAMS deliverables are submitted on time. After receiving RAMS deliverables, the review between the design engineer and SA member will be conducted and the result will be feed back to the concerned suppliers.

After reviewing Subcontractor's RAM analysis, it will be integrated into BEML's System level RAM analysis which will be used for arriving at vehicle level apportionment and RAM prediction value.

8. RAM Requirements

8.1. Reliability Requirements

The trains will achieve the following reliability requirement by the end of the warranty period:

1) The fleet average levels of MDBF, as specified in Table 1, shall be achieved.

Duration	Minimum fleet average MDBF
	6-Car fleet
After 6 months of start of revenue service	60,000
After 12 months of start of revenue service	100,000

Table 1: Reliability Target for Car Level

2) The achieved level of MDBCF of major systems shall be as follows.

S.N	System / Equipment	MDBCF (km) (Unit : 3car train-km)
(i)	Propulsion System	5,00,000
	a) Shoe collectors	53,00,000
	b) HSCB and Earthing Switch	13,50,000
	c) Traction Inverter	13,50,000
	d) Traction Motor	74,50,000
(ii)	Auxiliary Supply System	10,50,000
	a) Auxiliary Converter	18,00,000
	b) Battery Charger	34,00,000
	c) Back-up Battery	89,50,000
(iii)	Air Supply and Friction Brake Equipment	7,00,000
(iv)	Door System and Controls	10,00,000
(v)	VAC System	8,00,000
(vi)	Communication System	14,50,000
(vii)	Couplers and Draft Gear	69,52,000
	a) Automatic Couplers	124,00,000
	b) Semi Permanent Coupler	159,00,000
(viii)	Bogies	36,50,000
	a) Drive gear and coupling	83,00,000
	b) Primary Suspension	1,25,00,000
	c) Secondary Suspension	1,36,00,000
(ix)	Lighting System	74,50,000
(x)	TCMS	25,00,000

Table 2: MDBCF of major systems

Note: As the level of system reliability performance depends on the degrade level of equipment function, safety function and train operation polices/conditions, the MDBCF above will be subject to refinement in accordance with train withdrawal scenario to be determined during the final design stage.

8.2. Availability Requirements

The trains supplied shall achieved a minimum availability of 95 %

"Availability shall be assessed by the following measure:

$$\text{Percentage Availability} = 1 - \frac{[DT(OPM)+DT(CM)]}{\text{Total Time}} \times 100$$

Where:

(i) Total Time is the time in hours in the assessment period multiplied by the number of trains commissioned under the Contract.

(ii) DT(OPM), or Down Time due to Other Preventive Maintenance, is the total down time in hours due to Preventive Maintenance other than service checks, summed over all sessions carried out on all trains commissioned under the Contract during the assessment period. The trains shall not be due for major overhauls at the time of demonstration and such trains shall therefore be excluded from the assessment.

(iii) DT (CM), or Down Time due to Corrective Maintenance, is the total down time in hours due to corrective maintenance, summed over all sessions carried out on the trains commissioned under the Contract during the assessment period. Any unreasonable delay in handing-over the train for repairs for reasons not attributable to rolling stock shall be excluded. Time spent on train integrity inspections after train reformations arising from corrective maintenance work shall be included.

(iv) The down time DT(OPM) shall be counted starting from the moment when the train becomes unfit for service or work is physically started on a train, whichever is earlier, and shall end when the train is restored to service condition. If the train is withdrawn from revenue service specially for preventive maintenance, time spent on withdrawing the train and sending back the train to revenue service, if any, shall also be included.

(v) The down time DT(CM) shall be counted starting from the moment when the train becomes unfit for service or work is physically started on a train, whichever is earlier, and shall end when the train is restored to service condition. If the train is sent to

revenue service after the corrective maintenance, the time spent on sending back the train to revenue service, if any, shall also be included.

(vi) The down times DT (OPM) and DT (CM) shall also cover the full content of the maintenance work concerned, including safety precautions, inspections, servicing, replacement of equipment, defect detection and rectification, testing and restoration to service condition."

Availability Target: The trains supplied shall achieve a minimum availability of 95%. Detailed availability calculation are covered in Annexure-3.

The average availability of the trains shall be assessed after 18 months from the start of revenue operation with the first train supplied under the contract, in a specified train Maintenance Depot. The total maintenance down times for all trains shall be collected by the Engineer on monthly basis, and the average availability during the preceding six months, shall be worked out from the above formula.

In the event that the availability target is not achieved, the determination of availability achievement in the preceding six month period shall be continued at monthly intervals until the target is achieved.

In the event that the availability target is not achieved, the rolling stock shall, at his own expense, take whatever action is deemed necessary to meet the availability requirement.

8.3. Maintainability Requirements

The rolling stock will achieve the following general and quantitative maintainability requirements:

8.3.1. General Maintainability Requirement

(1) Simplicity of maintenance, operation and emergency procedures, ease of repair of damaged cars and equipment, are most important. These together with ease of exterior and interior cleaning have been taken into account throughout the development of the design.

(2) Particular attention is paid during the design of the cars to ensure that scheduled maintenance tasks are achieved in minimum time and using minimum manpower.

(3) Those components, systems and assemblies which require routine maintenance, frequent attention or unit replacement, shall be easily accessible for in situ maintenance.

(4) The BEML has developed a comprehensive maintenance programme for the trains.

(5) The maintenance regime proposed for the train was developed during the design process. A Failure Mode Effect Analysis (FMEA) is done based on function and derived from the specification at conceptual design stage.

(6) At pre-final design stage BEML developed this FMEA to include required maintenance derived from each failure mode. Any other maintenance required for the train should be indicated at this stage.

The vehicle shall incorporate design, which reduces maintenance, substantially improving service intervals and component replacement. The design shall also minimize mean time to repair (MTTR) and costs throughout design life. MTTR is the ratio of cumulative time, including the access time expended during a time interval to the total number of relevant failures.

The objective of the maintainability program including corrective and preventive maintenance shall provide for:

(i) Enhancement of Vehicle availability.

(ii) Minimisation of maintenance cost.

(iii) Minimisation of vehicle down time.

8.3.2. Quantitative Maintainability Requirement

The rolling stock will achieve the following maintainability targets:-

(i) LRU replacement takes less than 30 minutes

(ii) Rolling stock shall meet the MTTR of the identified key system in bid

S.N	System / Equipment	MTTR (Unit : Hour)
(i)	Propulsion System	2.5
	a) Third rail current collector	1.5
	b) HSCB and Earthing Switch	2.5
	c) Power converter - Inverter	3.0

	d) Traction Motor	3.0
(ii)	Auxiliary Supply System	1.7
	a) Auxiliary Converter	2.0
	b) Battery Charger	2.0
	c) Back-up Battery	1.0
(iii)	Air Supply and Friction Brake Equipment	2.0
(iv)	Door System and Controls	1.2
(v)	VAC System	1.5
(vi)	Communication System	1.2
(vii)	Couplers and Draft Gear	2.2
	a) Automatic Couplers	2.2
	b) Semi Permanent Coupler	2.0
(viii)	Bogies	2.0
	a) Drive gear and coupling	2.0
	b) Primary Suspension	2.0
	c) Mechanical Parts & Components	2.0
	c) Secondary Suspension	2.0
(ix)	Lighting System	1.0
(x)	TCMS	1.5

Table 3: MTTR of Major System

9. Methodology

9.1. Introduction

This section identifies the specific approaches adopted for each of the analyses and tasks carried out under the system assurance process.

All analysis defined in this chapter (e.g. Reliability Apportionment and Prediction.) except for Fault Tree Analysis (FTA) will be developed and compiled using the proprietary software, Microsoft Excel.

9.2. Reliability Apportionment and Prediction

9.2.1. Purpose

The purpose of Reliability Apportionment is to develop a model for presenting the functional interdependencies of the system/subsystem. It provides an analytical framework for apportioning the system level numerical requirements into lower level reliability estimates to generate design goals. The reliability prediction is a bottom up analytical process aims to estimate the system reliability performance and to determine whether the reliability targets can be achieved with the proposed design.

9.2.2. Procedure

The configuration of the system under review is represented diagrammatically by means of Reliability Block Diagrams (RBD), to be developer from the functional block diagrams, drawings and schematics. All equipment contributing to the system reliability performance, their redundancy arrangement as well as the success/failure conditions, are considered in the RBD development. In order to ensure that adequate attention have been given to all the system components and that their failure impacts are fully considered, it is important to perform bottom up analyses such as the FMECA prior to the construction of RBD.

Once the RBD is constructed, the system reliability requirements are allocated to the level specified and are used to establish the baseline requirements for designers. Requirements with the allocations are imposed on the sub-contractors. The equipment performance is estimated using past performance data, or engineering judgment if the data is not available, and be normalized to the design system application and operating environment. The system overall reliability performance will then be established based on the RBD model. Predications provide trade-off means among design alternatives.

It needs to be noted that reliability modeling, allocations and predications is a continuous and iterative process starting from initial system design and past failure rate information and ending with reliability demonstration.

9.2.3. Results

The RBD model allows for easy identification of the weak links in the functioning of a system in the context of its overall reliability performance. Allocations provide assurance that the total product will meet the design goals apportioned

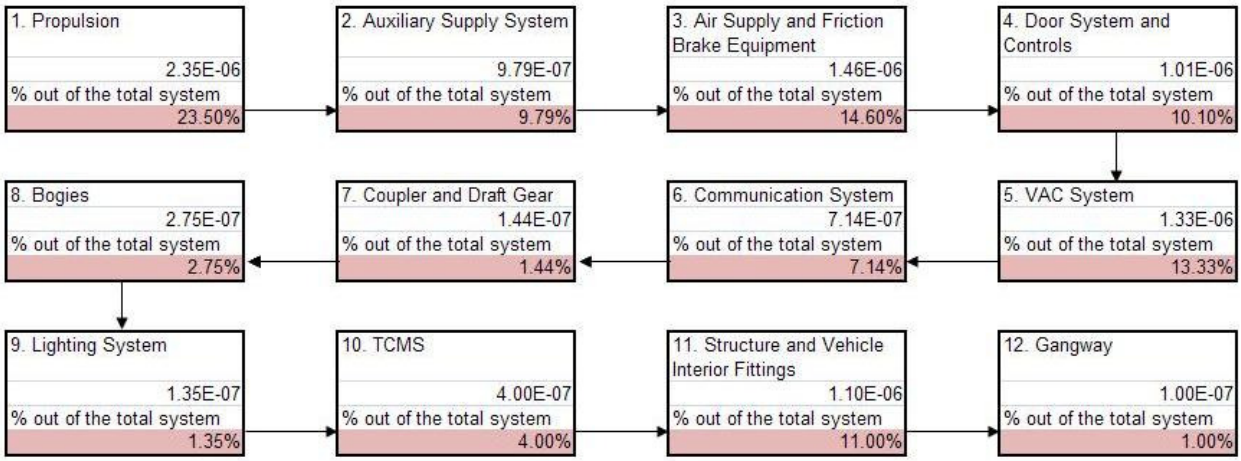
from the system requirements. The reliability predictions will provide quantitative evaluation of a design and highlight design areas having the largest potential for reliability improvement.

Reliability apportionment and Prediction has been carried out as indicated in Annexure-2. Failure rates of systems for 3 car unit (that are unique i.e., DMC, TC and MC) has been tabulated for apportionment. As the 6 car configuration may be assumed to be obtained by replication of two 3 car units (typical units), the Reliability apportionment and Prediction calculation can be considered to be same as all he characteristics remain the same except the direction of running.

9.2.4. Documentation

Reliability apportionment and prediction report is developed including Reliability Block Diagram (RBD) as below. RBD shows how the configuration of the system contributes to its overall reliability performance. Detailed system level reliability apportionments and reliability calculations are covered in the Annexure-1.

Reliability Block Diagram



	Failure Rate	MDBF
Overall Predicted Reliability Performance	1.00E-05	100,000
Overall Train Level Target	1.00E-05	100,000
% Within / Over Targets	100%	

Figure 4: Reliability Block Diagram (RBD)

9.3. Maintainability management and prediction

9.3.1. Purpose

The purpose of maintainability prediction is to develop a quantitative model for the vehicle, which can be used to estimate the MTTR for all individual systems and the overall vehicle.

9.3.2. Procedure

The maintainability assurance procedure outlines the general flow of tasks that will be followed to ensure that the rolling stock meets its maintainability requirements and satisfies all the requirements.

The general flow of maintainability tasks is illustrated in Figure 4.

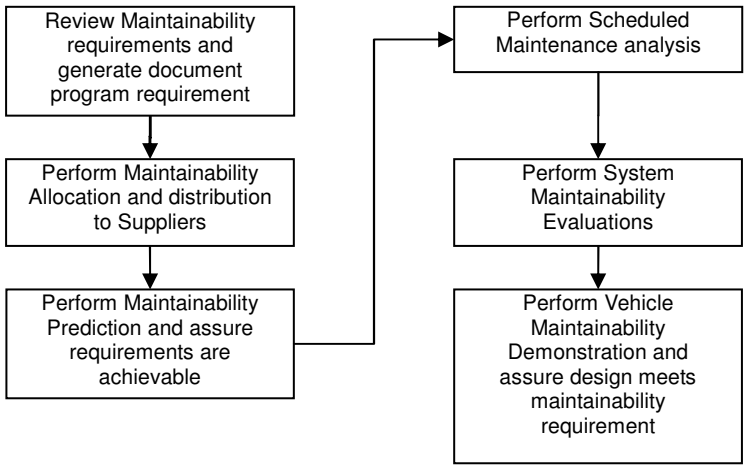


Figure 4. Maintainability Management Procedure

9.3.3. Results

The maintainability prediction provides a structured approach to evaluate the maintainability of the overall design and prioritize potential maintainability improvements. The maintainability prediction is a bottom up analytical process, which aims to estimate the system maintainability performance from LRU level maintenance tasks.

9.3.4. Documentation

Maintainability prediction results will be developed with a maintenance task evaluation at the LRU level to quantify the maintainability of each system and the vehicle overall as a part of reliability apportion and prediction report.

9.4. RAM Demonstrations

9.4.1. Purpose

The RAM Demonstration will be carried out during warranty period to provide KMRC with final assurance that the rolling stock meets its design objectives.

9.4.2. Procedure

The Reliability and Maintainability Demonstration will be carried out in accordance with the TS 19.3. and 19.7. During the warranty period, the reliability of the rolling stock will be assessed based on the results of the Failure Reporting and Corrective Action System (FRACAS). Failures due to incorrect maintenance, inadequate operating procedures, driver error, incorrectly performed procedures by e.g. station staff and failures of equipment other than that supplied under the contract are not considered in assessing rolling stock reliability performance.

9.4.3. Results

The results of the RAM Demonstration will provide an indication of the level of reliability, availability and maintainability which has been achieved by the rolling stock. This will allow comparison with the specific requirements for reliability and maintainability, and assessment of whether the rolling stock meets its objectives.

9.5. Failure Reporting and Corrective Action System (FRACAS)

During the defects liability period, the RAM performance of the rolling stock will be assessed based on the results of the Failure Reporting and Corrective Action System (FRACAS).

FRACAS will be set up to provide a documented history of any problems, failures, their associated corrective actions, as well as details on how and why each problem arose or failure occurred during the defects liability period. During the defects liability period, any unacceptable systematic weaknesses identified will be managed through design improvement and modification to the KMRC cars, in order to achieve the specified reliability targets.

Detailed FRACAS procedure is as follows;

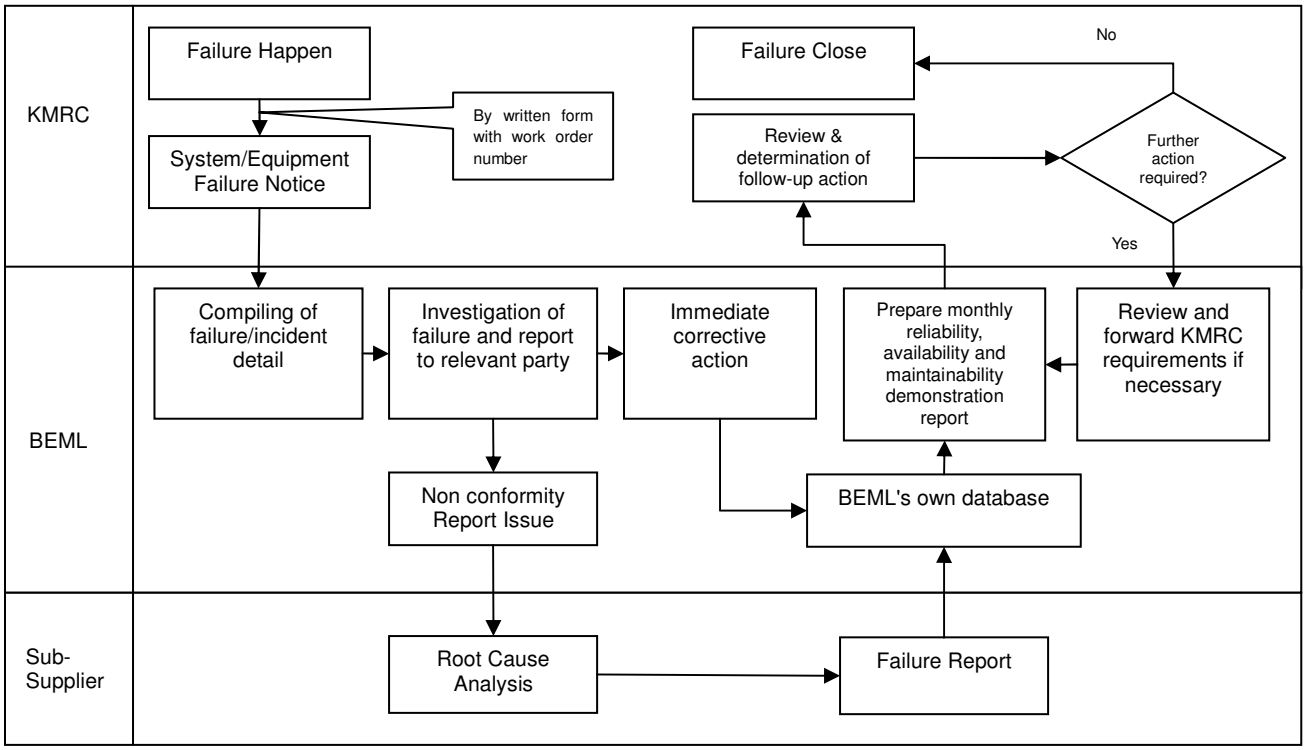


Figure 5: FRACAS Procedure

As part of the FRACAS, the following tasks will be undertaken in the RAM demonstration organization to review progress and co-ordinate design and reliability, availability and Maintainability activities.

- Review of the failure identified;
- Detection of failure causes from the demonstration results;
- Feedback of problems identified to relevant party and to the Performance Review Team;
- Determination of redesign, if required, based on review results; and
- Identification of deficiencies in design, material and workmanship through analysis on failure reports and trends.

The results of the RAM performance will provide an indication of the level of reliability, availability and Maintainability which has been achieved by the rolling stock. This will allow comparison with the specific requirements for RAM and assessment on whether the rolling stock meets its objectives.

Failures due to incorrect maintenance, inadequate operating procedures, and failures of equipment other than that supplied under the contract will not be considered in assessing rolling stock RAM performance.

10. Attachments:

Annexure-1: Reliability Block Diagrams

Annexure-2: Reliability Calculations

Annexure-3: Availability Calculations